Leiam Gaudin

317-556-0084 | leiamgaudin@gmail.com | linkedin.com/in/leiam-gaudin | Portfolio: tinyurl.com/lg-project-portfolio

Education

Purdue University West Lafayette, IN
Bachelor of Science in Mechanical Engineering *GPA: 3.52 | Expected May 2027*

Experience

Mechanical Engineering Intern May 2025 – Aug. 2025
Estes Design & Manufacturing *Indianapolis, IN*

- **Sheet Metal DFM:** Drafted **60+** engineering drawings in Creo; applied tolerance stack-up analysis to ensure assembly functionality for medical and defense enclosures.
- **Engineering Fundamentals:** Validated structural integrity for defense and medical assemblies through first-principles analysis, utilizing Free Body Diagrams (FBDs), beam deflection hand calculations, and stress/strain analysis to ensure safety factors met regulatory standards.
- **Tooling & Fixtures:** Saved company **$11,000** by designing 3D printed hardware fixtures and jigs using PETCF and threaded inserts; improved production tolerances by **57.1%** via parameter calibration.
- **Electromechanical Design:** Engineered a linear actuator and limit switch system for a **400+ lb** defense door; integrated wireless latch systems and performed Root Cause Analysis on industrial machinery.
- **Product Development:** Led R&D for medical spray chambers and secure defense lockers, reducing mechanical noise by **50%** through sound dampening integration.
- **Fabrication & Control:** Operated and performed Root Cause Analysis on Salvagnini L3 Laser and P4 Panel Bender systems with a detail-oriented approach; optimized nested layouts for sheet metal fabrication to minimize material waste and ensure grain alignment for critical bends.

Rapid Prototyping Lab Teaching Assistant Aug. 2025 – Present
Purdue University *West Lafayette, IN*

- **Design Consultation:** Provide DFM feedback for 20+ Senior Design teams, advising on part geometry, material selection, and slicer optimizations for complex physical builds.
- **Lab Management:** Manage a fleet of industrial-grade FDM/SLA printers, troubleshooting hardware failures and optimizing print settings for high-performance polymers.

Purdue FSAE Powertrain Member Aug. 2023 – May 2025
Purdue University *West Lafayette, IN*

- **Engine Simulation:** Utilized **GT-Power** for engine cycle simulation and developed **MATLAB** scripts for engine performance modeling to integrate with the team's lap time simulator.
- **Testing Hardware:** Designed and manufactured custom sensor mounts for the engine test stand, ensuring rigid alignment and data integrity under high vibration conditions.

Leadership

BattleBots: Railside Robotics | *Leadership, SolidWorks, MATLAB, 3D Printing* Sep. 2025 – Present

- Led the design and manufacture of a 1lb battlebot in a team of 4 students, integrating hardware and electronics.
- Developed impact optimized weapon geometry using a custom MATLAB center of mass optimization script, resulting in a **30%** increase in kinetic energy delivery for competition weapon blades.
- Utilized parametric modeling to optimize mass distribution, achieving **2x** more efficient mass usage for high stress drivetrain housings and internal component layouts.

VEX Robotics Competition | *Leadership, C++, PID, Mechatronics, Large Assembly CAD* Aug. 2021 – May 2024

- Led a team of 6 to design, build, and program an autonomous robot for the VEX v5 International Competition.
- Achieved top 50 global standing by managing engineering workflows across multiple competitive seasons.
- Implemented PID Control and bang bang loops for flywheel velocity management, incorporating laser distance sensors for anti-jam logic; ranked in the top **20%** globally.

Technical Skills

Design & Analysis: Sheet Metal DFM, Tooling/Fixture Design, GD&T, Tolerance Stack-up, FEA, Vehicle Dynamics, Drive Systems
Software: 3D CAD, Siemens NX, CATIA, SolidWorks, Creo, GT-Power, Simulink, ROS
Manufacturing: Rapid Prototyping (FDM/SLA), CNC Machining, Salvagnini L3 Laser/Bending
Programming: MATLAB, Python, C++, Arduino, Embedded Systems